

TYRA OEST, Denmark

WMO: 060220

Lat: 55.721N

Lon: 4.802E

Elev: 75

StdP: 100.43

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	0.2	-6.2	2.2	0.7	-5.2	2.5	1.1	25.1	5.2	23.5	5.2	11.4	100	1.040

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.7	19.0	17.3	18.2	16.7	17.9	16.4	17.7	18.5	17.1	18.1	16.5	17.6	7.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	17.9	16.8	12.1	17.7	16.0	11.5	17.3	50.2	18.9	48.2	17.8	46.4	17.6	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	20.9	19.0	-1.0	19.7	1.2	1.1	-1.9	20.5	-2.6	21.1	-3.3	21.8	-4.2	22.6		
(5)			-3.1	18.0	1.4	0.7	-4.1	18.5	-4.8	18.9	-5.6	19.3	-6.6	19.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	5.2	4.6	5.5	6.9	9.6	12.7	15.3	16.1	14.7	12.2	9.0	6.5		
	DBStd	4.47	2.16	2.09	1.85	1.79	2.07	1.73	1.65	1.41	1.61	1.80	2.37	2.11		
	HDD10.0	726	150	151	141	94	32	2	0	0	0	4	43	108		
	HDD18.3	3082	408	384	399	343	270	169	94	69	108	190	279	367		
	CDD10.0	687	0	0	0	1	20	82	166	191	142	72	14	0		
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.1	12.3	11.1	10.3	8.9	8.9	8.3	8.3	8.7	9.7	11.0	11.9	11.6		
(15) Precipitation	PrecAvg	1134	139	111	95	56	45	62	70	88	100	110	130	129		
	PrecMax	1404	244	185	166	99	69	108	101	145	165	158	205	251		
	PrecMin	918	49	49	33	24	23	24	33	37	46	44	81	63		
	PrecStd	113	48	38	31	17	13	18	20	24	28	31	31	43		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	8.2	9.0	11.8	16.1	18.0	20.1	20.1	19.0	16.2	14.2	11.1		
		MCWB	8.8	7.3	7.7	9.3	13.8	16.1	17.7	18.2	17.7	15.0	12.9	10.1		
	2%	DB	9.1	8.1	8.2	10.8	14.8	16.9	19.0	19.1	18.1	15.2	13.2	10.2		
		MCWB	8.1	7.2	7.2	8.7	13.1	15.4	17.1	17.4	16.6	13.9	11.9	9.2		
	5%	DB	8.2	7.8	7.9	10.0	13.2	15.9	18.1	18.2	17.2	15.1	12.8	9.9		
		MCWB	7.3	6.9	7.1	8.6	11.6	14.6	16.7	16.7	15.7	13.7	11.4	8.9		
	10%	DB	8.0	7.1	7.2	9.1	12.1	15.0	17.8	18.0	17.0	14.8	12.0	9.1		
		MCWB	7.0	6.3	6.4	8.3	10.6	13.8	16.4	16.4	15.5	13.4	10.5	8.2		
	0.4%	WB	8.9	7.7	7.7	10.1	15.0	16.4	18.1	19.1	17.7	15.5	13.5	10.6		
		MCDB	9.2	7.9	8.1	10.8	15.4	17.3	19.2	19.8	18.7	16.1	14.1	10.9		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	8.2	7.2	7.3	9.2	13.1	15.5	17.5	18.0	16.9	14.7	12.6	9.7		
		MCDB	8.6	7.6	7.7	9.9	14.4	16.6	18.6	18.6	17.8	15.4	13.4	10.0		
	5%	WB	7.6	6.8	6.9	8.7	12.0	14.8	16.8	17.3	16.3	14.3	11.6	9.0		
		MCDB	8.1	7.2	7.4	9.4	13.5	15.8	17.9	18.1	17.2	15.0	12.6	9.3		
	10%	WB	7.0	6.3	6.5	8.1	11.1	14.0	16.4	16.7	15.6	13.8	10.8	8.4		
		MCDB	7.6	6.8	7.1	8.9	12.2	15.1	17.5	17.7	16.6	14.5	11.9	8.9		
	5% DB	MDBR	2.0	1.8	1.8	1.9	1.9	1.8	1.8	1.7	1.7	1.8	2.1	2.1		
		MCDBR	2.3	1.8	2.0	2.3	3.4	2.7	2.4	2.0	1.7	1.8	2.3	2.3		
		MCWBR	2.8	1.8	1.8	1.7	2.5	1.9	1.8	2.1	2.0	2.2	2.8	2.9		
		MCWBR	2.2	1.7	1.9	2.1	3.5	2.3	2.2	1.9	1.5	1.9	2.1	1.9		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.345	0.364	0.396	0.378	0.384	0.388	0.390	0.376	0.355	0.334	0.312		
		taud	2.228	2.312	2.324	2.283	2.414	2.415	2.415	2.415	2.433	2.391	2.286	2.227		
	Ebn at Noon		557	715	797	822	864	862	849	823	780	697	554	487		
		Edh at Noon	58	77	96	115	107	108	106	100	86	71	55	46		
(44) All-Sky Solar Radiation	RadAvg		0.46	1.10	2.48	4.19	5.61	5.70	5.42	4.45	2.90	1.49	0.60	0.33		
	RadStd		0.06	0.17	0.23	0.38	0.43	0.41	0.53	0.34	0.32	0.14	0.08	0.04		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page